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Object-based video synopsis approach using particle swarm optimization

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Abstract

These days surveillance cameras are spreading very fast for security issues; however reviewing, retrieving and analyzing all these surveillance videos consume a lot of time. Video synopsis technology solved this problem by extracting all the active objects tubes that occurred at different times and relocating these objects simultaneously in a video for fast reviewing. Rearranging objects tubes in the video is considered the main challenge in creating video synopsis. Conventional methods proposed different approaches to optimize the energy function for relocating the object, but it suffers from high computational complexity and time-consuming. Moreover, some of these methods could not save the chronological order of moving objects. In this paper, the particle swarm algorithm is proposed for the first time to solve the energy minimization function and rearrange the objects to generate a condensed synopsis video with less collision and in chronological order. The experiments are applied to a benchmark dataset (VIRAT), and the preliminary simulation results demonstrate that the proposed method outperforms the genetic algorithm.

Keywords Video synopsis · Video surveillance · Particle swarm algorithm · Object detection and tracking · Deep SORT · YOLO

1 Introduction

Nowadays, a huge amount of cameras are used for security purposes 24 h a day. These cameras are producing an enormous amount of videos. Although the recorded videos contain useful information for security monitoring and criminal evidence applications, these videos have a lot of redundant information and analyzing these videos will be time-consuming [1]. Therefore, video synopsis is proposed as an intelligent solution that condenses the original video while conserving all the vital video activities. The synopsis video is generated by displaying all the objects simultaneously, even though they happened at different times. Video synopsis helps the user to watch long videos in just a few minutes or seconds, leading fast browsing, retrieval and analysis of videos without losing any crucial activity. Object-based video synopsis approach was proposed for the first time by Rav-Acha et al. [2]. First, they extract all the objects

tubes, where the tube is the representation of an object in space–time of all video frames, from the original video by object detection and tracking algorithms. Then, a condensed video is created by relocating all the objects tubes into a synthetic background based on an optimization method [3]. Therefore, the video synopsis approach mainly consists of four elements: background modelling, object detection, object tracking, and creating and optimizing the energy cost function. Although the video synopsis reduces the video duration, it may cause collisions between the objects, and this is considered one of the video synopsis challenges. Also, video synopsis approaches face other challenges such as losing chronological objects order and losing relation between objects. Researchers addressed these problems by proposing different methods. Li et al. [4] presented a video synopsis technique to decrease the collisions between objects tubes. The technique moves the objects in temporal domains, and if a collision has been detected, the size of objects is scaled-down. On the other hand, Nie et al. [5] proposed a video synopsis approach that shifts the objects in the time domain to generate the compressed video synopsis, and in spatial domains to reduce their collision. Lin et al. [6] proposed a distributed processing method to reduce the computational

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complexity of generating video synopsis. The input video is divided into different parts, and then each part is allocated to a computer to take the advantages of the multi-core capabilities. Tapas Badal [7] presented a video synopsis method to preserve the relations between objects. The method measures the interactions between objects by the difference between the objects tubes. If the difference is greater than a threshold, the tubes will be merged and produce a tube set. Also, Tian [8] proposed a trajectory mapping model to save the relationships of the objects and preserve the activities time consistency between the original and the synopsis videos. Yao et al. [9] found that the energy minimization function formula needs more computational time, so he proposed a new formula for the energy function and solved it using genetic algorithm (GA). Xu et al. [10] proposed an optimization problem based on GA to solve the temporal object tubes combination in creating video synopsis. The method proves that the GA outperforms the simulated annealing in terms of time consumption and information loss. Recently, the GA is compared to the particle swarm algorithm (PSO) [11], and it was found that PSO can reach the global minimum solution as GA but with less computational cost [12]. Therefore, in this paper, PSO is used to solve the energy minimization function and to generate the synopsis video with less collision and in chronological order.

The remaining sections of this paper are organized as follows. Section 2 presents a brief description of the video synopsis approach. Section 3 describes the proposed video synopsis approach. Section 4 illustrates the simulation results of the proposed methods. Finally, the conclusion is presented in Sect. 5.

2 Video synopsis background

Video synopsis is an efficient technology to compact a long video into a short one to ease the process of reviewing and analyzing the videos. Video synopsis generation is based on shifting and rearranging the objects in the temporal domain based on an optimization method. As shown in Fig. 1, the original video starts with a flying bird, and after a small inactive period, a person appears running in the street [13]. Video synopsis compacts the original video by using the area of the frames effectively and shifting the person from its frames to the bird frames for playing both of them simultaneously in less duration. Despite the simplicity of video synopsis approach, there are some issues that should be considered when moving and relocating the objects in the synopsis video: (1) The synopsis video should include all the objects and activities that appeared in the original video as much as possible. (2) The video synopsis length should be smaller than the input video. (3) The collision between objects should be avoided up to the hilt. (4) The chronological order of the

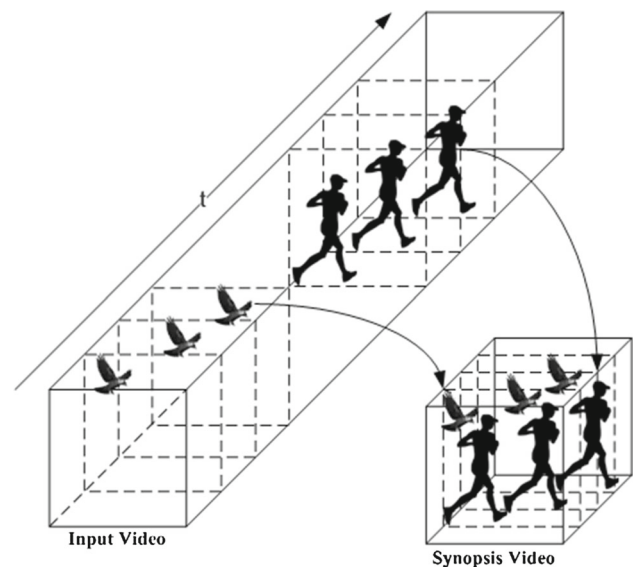


Fig. 1 Video Synopsis Idea

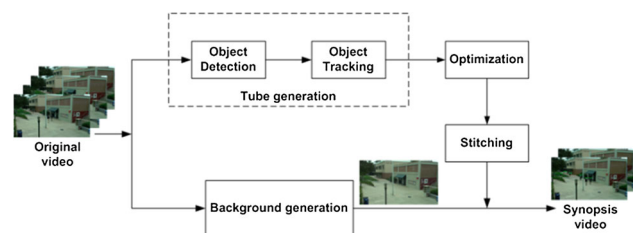


Fig. 2 Video synopsis framework

objects in both original video and synopsis video should be consistent.

Object-based video synopsis approach applied several processes on the original video to generate the synopsis video, as shown in Fig. 2. Tube generation is considered the first stage in the object-based approach, which generates a sequence of rectangles of the same object occurrence through the video frames. Tube generation depends on detecting and tracking the objects by linking the rectangle of the same object through video. Different methods have been suggested for detecting objects such as pixel difference with temporal median, background cut, Gaussian Mixture Models, frame difference, optical flow, adaptive background modelling and min-cut [14]. Meanwhile, for tracking these detected objects through the video frames to generate the object tubes, various tracking algorithms are proposed as multiple object tracking: Euclidian distance [15], 3D graph-cut [16], particle filter [17], Kalman filter [9]. Detection and tracking methods play an important role in the performance of the video synopsis because false detection and tracking affect the accuracy negatively.

The second stage (the most important step) determines the best location of the objects in the output video based on

the optimization method to present all objects in reduced time duration with minimum collision. The optimization method needs an objective function to minimize it; most of the research used the energy minimization function as an objective function to find the best location for the shifted objects with the lowest cost. The energy minimization function describes the interaction and the chronological order between the objects tubes. Therefore, the less the energy differences between the synopsis video and the original video, the higher the quality of the synopsis video [18].

Let the time duration of each tube b_o in the original video be $t_{b_o} = [t_{b_o}^s, t_{b_o}^e]$, where $t_{b_o}^s$ and $t_{b_o}^e$ are the start and end frames. The synopsis video is created by time mapping the objects tubes extracted from the original video $M(b_o) = b_s$ where M is the time shift. The new starting and ending frames of the object tubes in the synopsis video $t_{b_s} = [t_{b_s}^s, t_{b_s}^e]$ are determined by minimizing the following energy function [4]

$$E(M) = E_a(M) + \alpha E_c(M) + \gamma E_t(M) \quad (1)$$

where $E_a(M)$, $E_c(M)$ and $E_t(M)$ represent the activity cost, the collision cost and the temporal consistency cost, respectively. The activity cost term $E_a(M)$ used to guarantee that a maximum number of object observations are moved to the synopsis video. The collision cost $E_c(M)$ limits the collision of objects that happened due to shifting the objects tubes. However, the temporal consistency cost $E_t(M)$ maintains the relations and the chronological order of objects tubes to the highest level [18]. Several studies suggest different mathematical representations for each term in the energy equation [4, 18–20] and solve it using various optimization methods like simulated annealing, greedy optimization, mean shift, graph coloring approach, GA., etc.

Background generation is the third stage in video synopsis generation. This stage does not affect the compression efficiency of the synopsis video; it just improves its quality because it reduces the visual inconsistency. The most mentioned background generation method in the video synopsis techniques is the temporal median [5, 13, 21].

3 The proposed synopsis generation method

This section presents the proposed method for generating the synopsis video. As mentioned in Sect. 2, video synopsis generation has four stages. In the next subsection, the details of each proposed stage are explained.

3.1 The tube generation

In the tube generation stage, YOLOv3 (You Look Only Once) is used as an object detection method [22]. It is considered the newest YOLO version; it is based on Darknet-53 net-

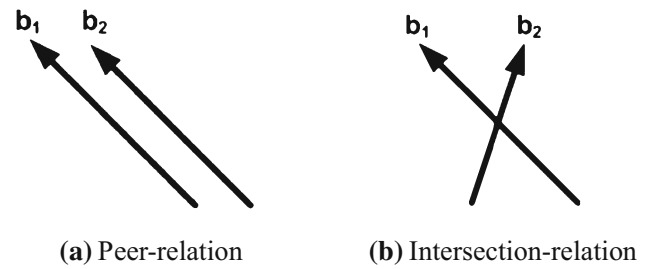


Fig. 3 Types of tubes relations

work and has 53 convolutions layers. YOLOv3 detects and determines the bounding box with high accuracy and also its processing time surpasses the faster R-CNN and RetinaNet models. After detecting and constructing the bounding boxes using YOLOv3, Deep SORT (Deep Simple Online Real-time Tracking) starts to track these bounding boxes through the whole video to generate the tubes [23]. Deep SORT tracking algorithm is applied to the video through 3 steps: Kalman filter, Hungarian algorithm and deep learning algorithm. The Kalman filter is responsible for motion estimation while the Hungarian algorithm is used for data association. The Deep learning algorithm is used to overcome the Kalman filter weakness in detecting occluded objects. The Deep learning algorithm takes the bounding box and returns a feature vector that describes the appearance of the object. Afterwards, a distance metric is used to measure the distance between those appearance vectors.

After generating the tubes, it is important to categorize the objects tube relations to save them in the synopsis video. From the previous studies, keeping the relation between moving objects tubes together is considered a challenge in video synopsis, since losing relations between objects may lead to losing important information as well as longer synopsis video. As shown in Fig. 3, there are two different types of tubes relations which are the peer relation (Fig. 3a) and the intersection relation (Fig. 3b). These relations are important, so they must be taken into account and reflected in the synopsis video.

- Tubes peer relation: as shown in Fig. 4, the proposed algorithm solves this case by merging the nearest tubes together after the tube generation stage. First, the proposed algorithm calculates the distance between the tubes according to the following equation:

$$d(b_i, b_j) = \begin{cases} d(b_i, b_j) & t_i \cap t_j > 0.75 * t_i \cup t_j \\ 0 & \text{Otherwise} \end{cases} \quad (2)$$

where t_i and t_j represent the presence time of tube b_i and b_j , respectively. $t_i \cap t_j$ is the intersecting time between tube b_i and b_j while $t_i \cup t_j$ is the total time of the two tubes. The proposed algorithm merges the tubes if the average

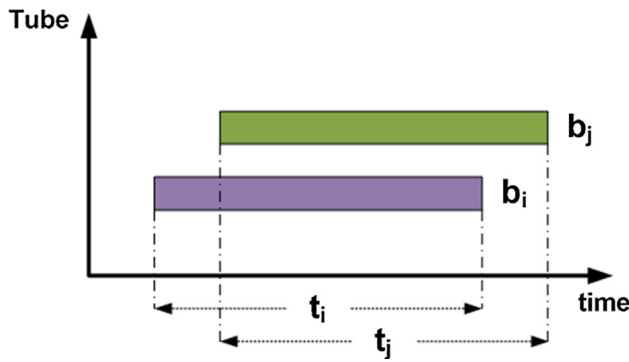


Fig. 4 Peer relation

value of the calculated distance $d(b_i, b_j)$ is less than the duration time of one of them.

- Tubes intersection relation: the optimization stage decides which intersection relations should be reserved based on their effect on the other synopsis parameters.

3.2 Optimization stage

In this stage, the extracted objects tubes are located in the synopsis based on PSO technique. In the proposed algorithm, the energy minimization is chosen to transfer all the moving objects from the original video to the synopsis video without changing their chronological order and avoiding collision as much as possible. The following steps describe how PSO is implemented to generate the synopsis video.

3.2.1 PSO parameters initialization

In this step, PSO parameters are defined since their values have a vital role in the performance and the efficiency of PSO. These parameters are the maximum number of iterations, the number of particles, personal and social acceleration coefficients, lower and upper bound of decision variables and inertia coefficient. Each particle is represented as a vector. Its length is equal to the number of tubes (n) and initialized by the starting time of each tube.

$$\text{Particle} = [t_{b1}^s \ t_{b2}^s \ t_{b3}^s \ t_{b4}^s \ \dots \ t_{bn}^s] \quad (3)$$

3.2.2 Energy minimization function

As mentioned in Sect. 2, the energy minimization function has three terms: activity cost, collision cost and temporal consistency. In the proposed method, all the extracted objects are mapped to the synopsis video; therefore, the activity cost term in the energy function is set to zero. Besides the collision cost and the temporal consistency, the proposed algorithm includes other aspects to the energy minimization function to increase the performance of the video synopsis. Let b rep-

resents a tube in the original video and b_s is the mapped tube in the synopsis video. The proposed energy minimization function is as follows:

$$E(M) = \alpha E_c(M) + \gamma E_t(M) + \kappa E_L(M) + \omega E_{tc}(M) \quad (4)$$

where E_c represents the collision cost, E_t is the temporal consistency cost, E_L is the synopsis length cost, and E_{tc} is the true collision cost. α , γ , κ , ω are collision weights, temporal consistency weight, synopsis length-weight and true collision weight, respectively. If one of these weights is assigned to a high value, this means that this term will be significant over the other terms. Each term of the energy function is described in details as follows:

- Collision cost

This energy term calculates the collisions between the tubes that happened in the synopsis video due to the mapping process.

$$E_c = \sum_{b, \hat{b} \in s} (b_s \cap \hat{b}_s) \quad (5)$$

where $b_s, \hat{b}_s$ are two tubes in the synopsis video.

- Temporal consistency cost

This energy term preserves the chronological order between the objects as much as possible. In this term, each tube is assigned an order according to its appearance in the video; then the difference between these tubes orders in the original and the synopsis is computed.

$$E_t = \sum_{b_o, b_s \in T} |\text{order}(b_o) - \text{order}(b_s)| \quad (6)$$

where T represents the set of all the tubes.

- The synopsis length cost

This term controls the length of the synopsis, so it does not exceed the longest tube length as much as possible. In the proposed method, segmenting long object tubes into different segments is avoided because this changes the chronological order of the objects and leads to displeasing visual synopsis video [5, 24].

$$E_L = \text{length}(s) - \text{Max}(|t_{b_s}^e - t_{b_s}^s|) \quad (7)$$

- True collision cost

True collision cost is responsible for preserving the intersection relation that occurred in the original video to be found in the synopsis video. The true collision cost is calculated as the difference between the number of true intersections that took place in the original video and the number of reserved true intersections in the synopsis video.

$$E_{tc}(M) = N_{TI} - N_{STI} \quad (8)$$

where N_{TI} is the number of true intersections, while N_{STI} is the number of reserved true intersections in the synopsis video. The proposed method calculates the number of the true collisions that occurred in the original video by counting the number of intersecting tubes as:

$$N_{TI} = \begin{cases} N_{TI} + 1 & t_i \cap t_j > 0 \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

The number of reserved true intersections is computed by comparing the time interval between the starting of the two intersecting tubes in both original and synopsis videos.

$$N_{STI} = \begin{cases} N_{STI} + 1 & t_i^s - t_j^s|_o = t_i^s - t_j^s|_s \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

In the end, the whole optimization function is:

$$E = \min_t \left(\alpha \sum_{b, \hat{b} \in s} (b_s \cap \hat{b}_s) \right) + \gamma \sum_{b_o, b_s \in T} |\text{order}(b_o) - \text{order}(b_s)| \\ + \kappa (\text{length}(s) - \text{Max} |t_{b_s}^e - t_{b_s}^s|) + \omega (N_{tc} - N_{stc}) \quad (11)$$

PSO optimizes this energy function to find the best starting time for each object tube to generate a short video synopsis with minimum collision and maximum chronological order. In most cases, all of these cost functions cannot be accomplished together; thus, the PSO has to make a trade-off between them. Additionally, the user can control the importance of these objectives by changing the weight given to each one of them. Increasing the weight of a specific term reflects the significance of this term for the user.

3.3 Background generation and objects stitching

In this stage, the background is generated based on the temporal median method, which is mentioned in Sect. 2. For stitching the objects tubes in the generated background, the Poisson method is used. However, to make the image more natural after stitching the objects, a mask is created for each object in the bounding box based on some morphological operations to get the objects without the surrounding background in the box. Then the seamless cloning based on the Poisson method stitches these extracted objects with the generated background.

4 Experimental results

This section presents and evaluates the proposed synopsis algorithm performance. Up to now, video synopsis research suffers from missing global metrics to evaluate the synopsis algorithms performance. Therefore, the most common metric which is the Condensation Ratio (CR) is chosen in this research to evaluate the synopsis performance. CR is defined as the ratio of the number of frames in the source video to the number of the frames in the output synopsis video [24]. Eleven videos are used to evaluate the performance of the proposed algorithm, 9 videos (vid1: vid9) from VIRAT Release 2.0 captured by stationary HD cameras (1080p or 720p) [25] while the other 2 videos (vid10: vid11) are downloaded from YouTube.

First, the visual performance of merging the peer-related objects tubes is presented in Fig. 5 using vid2. As shown in Fig. 5b, if the related tubes are not merged, the PSO starts each tube at different time. Thus, the relation between any two persons moving beside each other will be lost, as well as increasing the synopsis video length. Also, if two persons are moving beside each other and they are holding hands as in Fig. 5b their hands are separated, which distorts the synopsis video visually. The proposed technique solves this problem by merging the related tubes to a tube set and deal with it as a single tube, as shown in Fig. 5c. This improves the synopsis video visually and reduces its length.

Second, the PSO parameters are initialized as maximum number of iterations = 300, number of particles = 50, lower decision bound = 0, upper decision bound = 10, personal acceleration coefficient 1 = 1.4962, social acceleration coefficient 2 = 1.4962 and inertia coefficient = 0.7298. Also, the values of the energy weights are set as $\alpha = 0.6$, $\gamma = 0.05$, $\kappa = 0.05$, $\omega = 0.3$. The weight assigned to each energy term controls its priority versus the other weights. The performance of the synopsis video is evaluated based on moderate weight values, not biased to any energy term.

As well, the proposed method based on PSO is compared to GA in terms of synopsis length, total collision area and the number of reserved intersections and the results are tabulated in Table 1. The GA parameters values are initialized as the PSO's parameters which are the maximum number of iterations = 300 and the number of chromosome = 50. However, the crossover probability and the mutation probability are initialized as $P_c = 0.9$ and $P_m = 0.01$, respectively, according to [10].

It can be noticed from the table that both techniques attempted to keep the synopsis length equal to the longest tube while reserving the other synopsis parameters, such as the collision. However, it can be observed that PSO outperforms GA for all videos in terms of collision values. On the other hand, for vid3, vid4 and vid7, GA reduces its synopsis length while increasing the collision between objects.

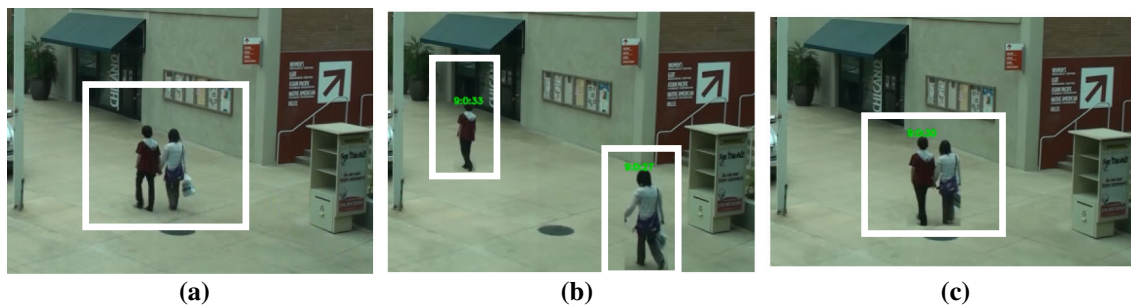


Fig. 5 Tube merging effect on the synopsis video **a** original video. **b** Video synopsis without tube merging. **c** Video synopsis with tube merging

Table 1 The performance of the proposed video synopsis technique

Video	Resolution	No of Frames									
		Original video	Longest tube	Synopsis Length		Collision		Condensation ratio		Reserved intersections	
				PSO	GA	PSO	GA	PSO	GA	PSO	GA
Vid1	1280 × 720	589	283	284	287	0.00	0.00	0.482	0.487	0 of 0	0 of 0
Vid2	1280 × 720	1669	1115	1316	1363	22.82	61.82	0.788	0.817	1 of 6	0 of 6
Vid3	1280 × 720	1938	585	963	741	28.99	122.97	0.497	0.382	2 of 6	0 of 6
Vid4	1280 × 720	776	594	667	653	0.08	0.12	0.86	0.841	0 of 0	0 of 0
Vid5	1280 × 720	1087	333	334	342	3.59	3.8	0.307	0.315	0 of 0	0 of 0
Vid6	1280 × 720	1904	1110	1110	1125	6.22	16.79	0.583	0.591	0 of 2	0 of 2
Vid7	1280 × 720	1020	916	1017	992	13.51	28.34	0.997	0.973	1 of 12	0 of 12
Vid8	1280 × 720	560	443	444	444	0.00	0.00	0.793	0.793	1 of 1	1 of 1
Vid9	1280 × 720	1007	963	963	990	0.16	38.1	0.956	0.983	1 of 1	0 of 1
Vid10	960 × 720	985	120	120	120	0.00	0.00	0.122	0.122	0 of 0	0 of 0
Vid11	1280 × 720	270	210	212	210	0.19	2.3	0.785	0.778	0 of 0	0 of 0

Increasing the collision affects the visual quality of the synopsis video severely, which is not preferred.

In this study, the synopsis length–weight is increased to reduce the synopsis length for the proposed method for vid2, vid3 and vid7, as shown in Fig. 6. The synopsis length in Fig. 6 is calculated as the difference between the synopsis length and the longest tube. It was found that increasing the synopsis weight decreases the synopsis length, but in contrast, the collision value increases, as shown in Fig. 7. So moderate weights are selected to assure the balance between the various parameters.

Moreover, from the point of view of the condensation ratio, it was mentioned previously that the value of the condensation ratio is affected by the synopsis length which should be equal to the longest tube in the original video. And this justifies the high condensation ratio in Table 1 for the videos that their longest tube approximately equal to the original video.

Furthermore, the computational complexity of both techniques is computed for the same number of iterations and the number of particles for PSO (chromosome for GA). All

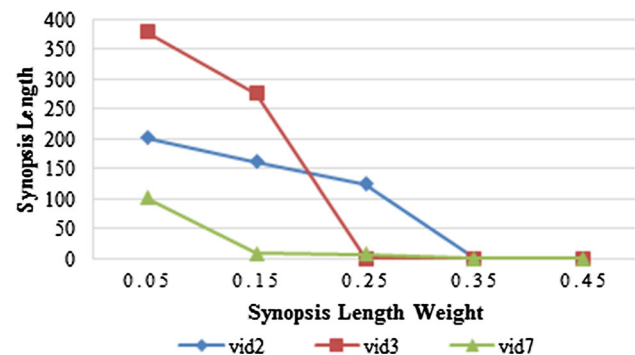


Fig. 6 Effect of the length–weight on the synopsis length

experiments have been carried out in Python 3.6 on a 2.5 GHz i5 Laptop computer. Figure 8 shows the consuming time for the 11 videos for both PSO and GA. It can be perceived that PSO consumes less time for all the videos by approximately 1.5 to 138 s.

Additionally, this study compared between GA and PSO in terms of true intersection relations that occurred in the original video and reserved in the synopsis video. It can be

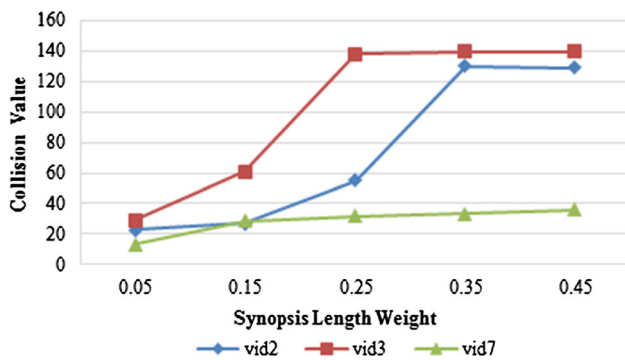


Fig. 7 Effect of synopsis length-weight on the collision

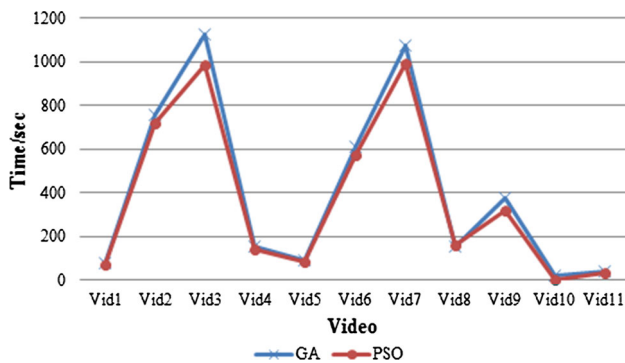


Fig. 8 Comparison of time consumption between PSO and GA

observed from Table 1 that PSO reserved some true intersections in vid2, vid3, vid7, and vid9 while GA could not reserve any of them. Figure 9b shows the output synopsis video from the proposed method for vid3 and vid6 if the true collisions energy term is omitted from the equation. While Fig. 9c shows the output video synopsis if the true collision is taken into consideration in the energy function. Reserving more intersection relations requires increasing the weight of

Table 2 Effect of true collision weight on the length of the synopsis video

Video	Number of intersections	No of Frames	
		$\omega = \text{Low}$	$\omega = \text{High}$
Vid2	6	1318	1524
Vid3	6	969	1147
Vid6	2	1110	1315
Vid7	12	964	1014

the true collision to give the true collision energy term a high priority. The result of increasing the true collision weight is presented in Table 2. It can be inferred from the table that increasing the true collision weight reserves all the intersection relations. However, this leads to an increase in the number of frames for synopsis video.

5 Conclusion

An object-based video synopsis approach based on particle swarm optimization (PSO) is proposed to alleviate the efforts of browsing long surveillance videos. To the best knowledge of the authors of this paper, this is the first time for the PSO to be used in video synopsis applications. The proposed method suggests an energy function that creates shorter video synopsis with avoiding collision among objects, keeping their peer and intersection relations and saving their chronological orders. The proposed method based on the PSO is compared to the GA algorithm. The preliminary simulation results of undergoing research demonstrated that PSO outperforms GA in terms of collision and length. The proposed method also succeeds in keeping the peer relation between the objects

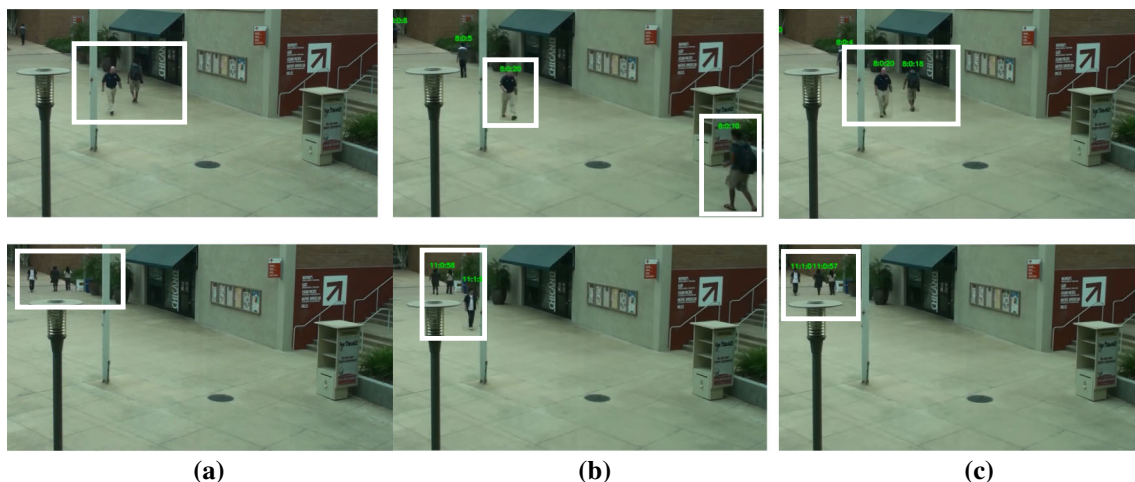


Fig. 9 Intersection relation effect on synopsis video **a** Original video (row1 = vid3 and row2 = vid6). **b** Synopsis video without reserving intersection relation. **c** Synopsis video with reserving intersection relation

tubes and saves intersection relations as much as possible. Also, it can be concluded from the results that PSO converges faster than GA to find the minimum of the energy function, which leads to less consumption time.

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